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Community Detection on Signed Graphs: A Bayesian MCMC Coupling Approach

Seminar of the MathNet team (Inria Paris)

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Outline

1. Motivation & the Sankararaman–Baccelli Model
2. The Bayesian Framework & Coupling
3. MCMC Dynamics & Performance Bounds
4. Open Questions & Conclusion

01
Motivation & the Sankararaman–Baccelli Model

Haplotype Assembly

1. Biological Setup: Two Homologous Chromosomes (Haplotypes)

Humans are diploid: they have two copies of each chromosome (homologous chromosomes).

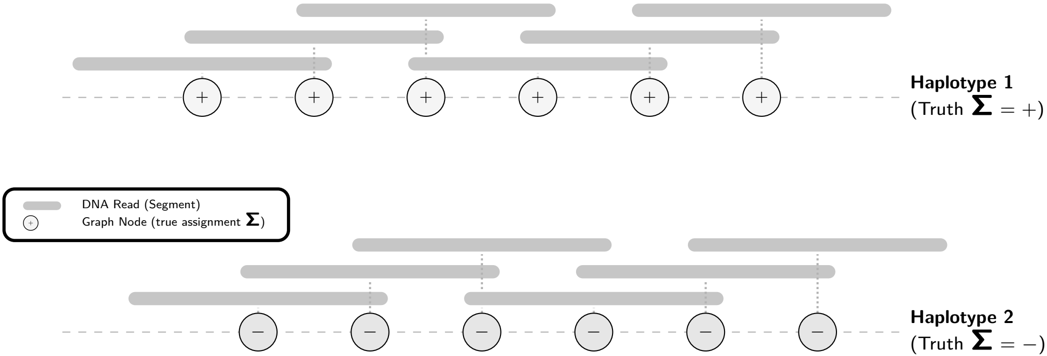
Haplotype 1
(Truth $\boldsymbol{\Sigma} = +$)

Haplotype 2
(Truth $\boldsymbol{\Sigma} = -$)

Haplotype Assembly

2. High-Throughput Sequencing: DNA Reads

Sequencing generates short, overlapping fragments called **reads** from both chromosomes.

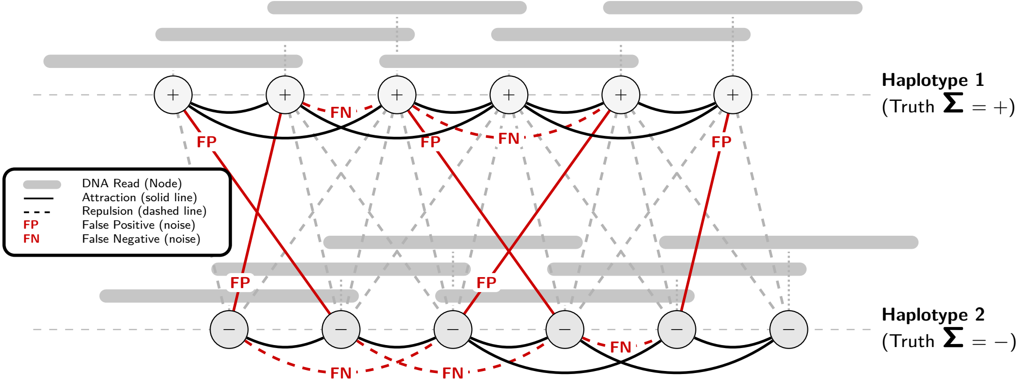


Haplotype Assembly

3. Modeling as a Signed Graph

Nodes = reads   Edges = overlaps

Goal: Reconstruct the true haplotype assignments $\boldsymbol{\Sigma}$ from this observed signed graph.



Motivation: Signed Community Detection

Community Detection on Signed Graphs ($K = 2$)

- We aim to partition nodes into $K = 2$ communities: $\boldsymbol{\Sigma}_i \in \{+1, -1\}$.
- Observed links represent **attractions** (solid lines) and **repulsions** (dashed lines).
- Under the hood, this is a **Stochastic Block Model** (SBM) setup.

The Geometric Challenge

- Nodes are embedded in space (e.g., genomic positions on $\mathbb{Z}$ or $\mathbb{R}^d$).
- Edge probabilities depend directly on the distance $r = \|x_i - x_j\|$:

$$\mathbb{P}(\text{attractive edge} \mid \text{same community}) = f_{\mathrm{in}}(r)$$
$$\mathbb{P}(\text{repulsive edge} \mid \text{different communities}) = f_{\mathrm{out}}(r)$$

- **All the difficulty (and the interest!) comes from this geometric dependency.**

The Sankararaman–Baccelli Model (GSBM)

Model Definition [1]

- **Spatial Nodes** V_n: Hom. Poisson PP $\mathcal{X}$ in $\mathbb{R}^d$ [2] restrict. to a n-volume window.
- **Latent Communities**: Uniform assignments $\Sigma_i \in \{+1, -1\}$ $(K = 2)$.
- **Geometric Overlaps**: Observed connection $e = \{i, j\} \in H$ with probability:

$$\mathbb{P}(e \in H \mid X, \Sigma) = \begin{cases} f_{\mathrm{in}}(r_e) & \text{if } \Sigma_i = \Sigma_j \text{ (attraction)}, \\ f_{\mathrm{out}}(r_e) & \text{if } \Sigma_i \neq \Sigma_j \text{ (repulsion)}, \end{cases}$$

with distance $r_e = \|X_i - X_j\|$ and connecting functions $f_{\mathrm{in}}(r) \geq f_{\mathrm{out}}(r)$.

Goal: Weak Recovery

- Find an algorithm τ (independent of Σ) reconstructing assignments:

$$\lim_{n\to\infty} \mathbb{E}[\mathrm{ov}_n(\Sigma, \tau)] \geq \frac{1}{2} + \eta \quad \text{for some } \eta > 0,$$

using the **Community Overlap**: $\mathrm{ov}_n(\Sigma, \tau) = \max_{\pi \in \mathfrak{S}_2} \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}_{\{\tau_i = \pi(\Sigma_i)\}}$.

Gibbs Model on Signed Graphs

Attractive and Repulsive Edges

We partition the potential edges $E = F \dot{\cup} \tilde{F}$:

- F: **Attractive** (ferromagnetic) edges, with weight $W_e > 0$
- $\tilde{F}$: **Repulsive** (antiferromagnetic) edges, with weight $W_e < 0$

Signed Gibbs Measure [3]

The probability of a label configuration $\sigma \in \{+1, -1\}^{V_n}$ is $\mu(\sigma) = \frac{1}{Z} e^{-U(\sigma)}$, where the energy $U(\sigma)$ penalizes unsatisfied edges:

$$U(\sigma) = \sum_{e=\{i,j\} \in F} W_e \mathbb{I}_{\{\sigma_i \neq \sigma_j\}} + \sum_{e=\{i,j\} \in \tilde{F}} |W_e| \mathbb{I}_{\{\sigma_i = \sigma_j\}}$$

The magnitudes $|W_e|$ determine the strength of each constraint.

From GSBM to the Gibbs Posterior

Conditioning on the Point Cloud

- With fixed spatial positions X, all pairwise distances $r_e = \|X_i - X_j\|$ are known.
- Under a uniform prior on Σ, the **posterior** distribution given the observed graph H is:

$$\mathbb{P}(\Sigma = \sigma \mid H, X) \propto \mathbb{P}(H \mid X, \sigma)$$

Log-Likelihood as Signed Gibbs Energy

- Choosing weights $W_e(H)$ as below:

$$W_e(H) = \begin{cases} \log\left(\dfrac{f_{\mathrm{in}}(r_e)}{f_{\mathrm{out}}(r_e)}\right) > 0 & \text{if } e \in H \text{ (attractive } \in F\text{)}, \\ -\log\left(\dfrac{1 - f_{\mathrm{out}}(r_e)}{1 - f_{\mathrm{in}}(r_e)}\right) < 0 & \text{if } e \notin H \text{ (repulsive } \in \tilde{F}\text{)}. \end{cases}$$

- ... yields the exact correspondence:

$$\mathbb{P}(H \mid X, \sigma) \propto e^{-U(\sigma)}$$

From GSBM to the Gibbs Posterior

Conditioning on the Point Cloud

- With fixed spatial positions X, all pairwise distances $r_e = \|X_i - X_j\|$ are known.
- Under a uniform prior on Σ, the **posterior** distribution given the observed graph H is:

$$\boldsymbol{\mathbb{P}(\Sigma = \sigma \mid H, X) = \mu(\sigma)} \propto \mathbb{P}(H \mid X, \sigma)$$

Log-Likelihood as Signed Gibbs Energy

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- ... yields the exact correspondence:

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The Percolation Obstruction

Necessary Condition for Weak Recovery [1]

- Fragmented graphs (small, isolated components) prevent global reconstruction.
- Global information flow requires the graph to **percolate** [4] (possess a giant component).

Theorem (Sankararaman & Baccelli [1])

Independent bond percolation with edge probability

$$p_e = f_{\mathrm{in}}(r_e) - f_{\mathrm{out}}(r_e)$$

yielding a giant component is a ***necessary condition*** *for weak recovery.*

- **Our goal**: Retrieve and extend this result using unified **Bayesian MCMC coupling**.

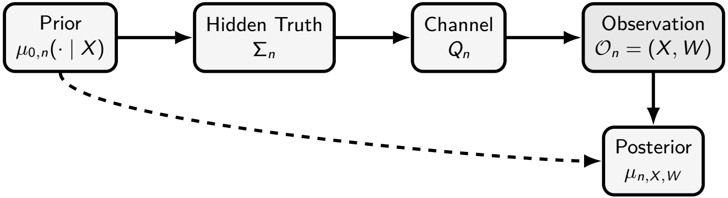
02
The Bayesian Framework & Coupling

The General Bayesian Model

Definition

1. **Nodes, Edges, Communities**: Nodes $V_n = \{1, \dots, n\}$, edges E_n, communities $\mathcal{L}_K = \{0, \dots, K-1\}$, and set of configurations $\sigma \in \mathcal{C}_n = (\mathcal{L}_K)^{V_n}$.
2. **Features**: $X_n \in \mathcal{X}_n$ according to distribution ν_n.
3. **Prior on configurations**: $\mu_{0,n}(\mathrm{d}\sigma \mid x)$ over $\mathcal{C}_n$.
4. **(Noisy) channel**: $Q_n(\mathrm{d}w \mid x, \sigma) = \prod_{e \in E_n} Q_{e,n}(\mathrm{d}w_e \mid x_i, x_j, \sigma_i, \sigma_j)$.

The joint law is $\mathbb{P}_n(\mathrm{d}x, \mathrm{d}\sigma, \mathrm{d}w) = \nu_n(\mathrm{d}x)\mu_{0,n}(\mathrm{d}\sigma \mid x)Q_n(\mathrm{d}w \mid x, \sigma)$.



Posterior Resampling (Nishimori Identity)

Replacing Truth by a Replica

- Comparing the algorithm's output with the hidden truth Σ_n is equivalent to comparing it with a posterior replica $\Sigma_n^{(1)}$.

Theorem (Posterior Resampling / Nishimori Identity [5])

Let τ_n be any community detection algorithm. Let $\Sigma_n^{(1)} \sim \mu_{n,X_n,W_n}$ be a replica drawn independently from the posterior. For any bounded test function Ψ_n:

$$\mathbb{E}\left[\Psi_n(X_n, W_n, \Sigma_n, \tau_n)\right] = \mathbb{E}\left[\Psi_n(X_n, W_n, \Sigma_n^{(1)}, \tau_n)\right].$$

Corollary (Overlap Identity)

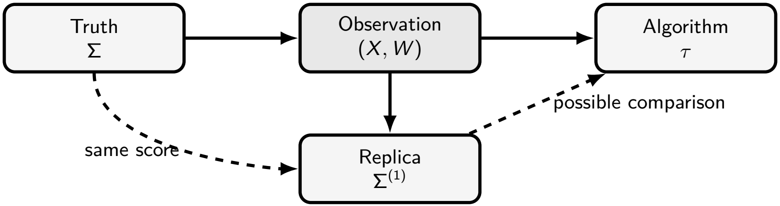
For any algorithm τ_n and threshold $s \in [0, 1]$:

$$\mathbb{P}\left[\mathrm{ov}_n(\Sigma_n, \tau_n) \geq s\right] = \mathbb{P}\left[\mathrm{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq s\right].$$

Invariant Coupling

Markov Chain Dynamics as a Source of Coupling

- We construct the posterior replica $\Sigma^{(1)}$ starting from the hidden truth Σ.
- Running **exactly one step** of a posterior-invariant $\textsc{Markov}$ kernel M initialized at Σ yields a valid replica $\Sigma^{(1)} \sim M(\Sigma, \cdot)$.

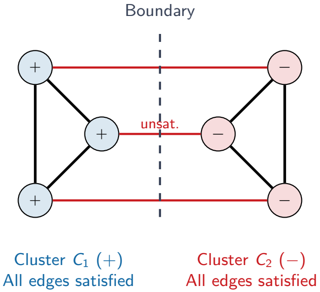


03
MCMC Dynamics & Performance Bounds

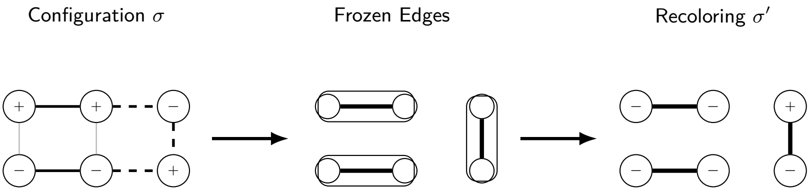
Single-Spin Dynamics & Geometric Bottleneck

Single-Spin Update Limitations

- **Geometry makes updates slow**: Single-spin dynamics (Glauber / Metropolis–Hastings [6]) update one node at a time.
- **Contradictory Clusters**: Two internally valid (highly satisfied) but mutually contradictory clusters are locked.
- **High Energy Barrier**: Flipping a single boundary node violates internal edges, creating a local energy barrier.
- **Conclusion**: Need for geometry $\implies$ **Cluster Dynamics** (updating whole regions).



One Step of Signed Swendsen–Wang [7, 8]



- Edges in bold represent satisfied interactions that are randomly frozen.
- Clusters are then flipped independently.

Swendsen–Wang Signed Cluster Dynamics

Swendsen–Wang (SW) Signed Cluster Dynamics [7, 8]

- Update whole clusters of vertices simultaneously to speed up mixing.
- **Freezing Step**: For each edge $e = \{i, j\}$:
  - If e is satisfied ($W_e > 0$ and $\sigma_i = \sigma_j$, or $W_e < 0$ and $\sigma_i \neq \sigma_j$), freeze it with probability:

$$p_e^{\text{frozen}} = 1 - e^{-|W_e|}.$$

  - If e is unsatisfied, do not freeze it ($p_e^{\text{frozen}} = 0$).
- **Recoloring Step**: Flip each frozen cluster independently with probability $1/2$.

High-Order Triangular Dynamics

The Frustration Obstruction

- **Frustration**: $\nexists \sigma$ satisfying all edges (conflicting constraints).
- **Excessive Freezing**: In frustrated configurations, standard SW satisfies and freezes too many edges independently, blocking/freezing the MCMC dynamics.
- **Higher-Order Correlation**: Group interactions into **triangles** to correlate freezing and **freeze as little as possible**.

Triangular Cluster Dynamics [9]

- **Energy Decomposition**: Distribute edge energy U over triangles: $U(\sigma) = \sum_{t \in \mathcal{T}} U_t(\sigma)$.
- **Freezing Mechanism**: Apply a joint freezing mechanism on the triangles (rather than independent edges) to keep cluster sizes active.

Triangular Dynamics Freezing Rules

$\triangle$	**Attractive Triangle** (Isotropic)				
Configuration	$\varnothing$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+,+,+)$	b_u	0	0	0	a_u
Other	1	0	0	0	0
$\triangle$ (dashed)	**Repulsive Triangle** (Isotropic)				
Configuration	$\varnothing$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+,+,+)$	1	0	0	0	0
$(+,+,-)$	b_u	0	$a_u/2$	$a_u/2$	0
$(+,-,+)$	b_u	$a_u/2$	0	$a_u/2$	0
$(+,-,-)$	b_u	$a_u/2$	$a_u/2$	0	0

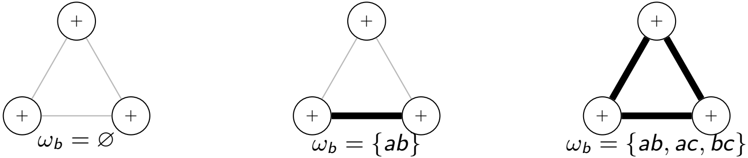
Table: Freezing rules for triangular dynamics ($a_u = 1 - e^{-2u}$, $b_u = e^{-2u}$), where u is the energy transferred by each edge.

General Swendsen–Wang Cluster Dynamics

General Framework [10]

Suppose the Hamiltonian decomposes as: $U(\sigma) = U_0(\sigma) + \sum_{b \in \mathcal{B}} U_b(\sigma)$, where each $b \in \mathcal{B}$ is a **bond** (e.g. edge, triangle, clique).

1. **Freezing Step**: For each bond $b \in \mathcal{B}$, freeze a sub-bond $\omega_b \subseteq b$ with probability $P_b^\sigma(\omega_b)$.
2. **Recoloring Step**: Recolor each frozen cluster independently according to a distribution preserving the prior $\mu_0(\sigma) \propto e^{-U_0(\sigma)}$.



General Swendsen–Wang Cluster Dynamics

Definition (Local Reversibility)

For two configurations σ, σ' and a bond b, we define the compatibility probability:

$$P_b^{\sigma \to \sigma'}(\omega_b) = \begin{cases} P_b^{\sigma}(\omega_b), & \text{if } P_b^{\sigma'}(\omega_b) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The freezing rule is **locally reversible** if for all b, ω_b, σ, σ':

$$P_b^{\sigma \to \sigma'}(\omega_b) = e^{-(U_b(\sigma') - U_b(\sigma))} P_b^{\sigma' \to \sigma}(\omega_b)$$

Theorem (Gibbs Invariance [10])

If freezing choices are independent, locally reversible, and $P_b^{\sigma}(\varnothing) > 0$ for all b, σ:

1. *The global cluster dynamics is reversible with respect to $\mu(\sigma) \propto e^{-U(\sigma)}$.*
2. *The condition $P_b^{\sigma}(\varnothing) > 0$ ensures the **irreducibility** of the Markov chain.*

Percolation as a Necessary Condition for Recovery

Theorem (Percolation Bound on Overlap [10])

Under the uniform prior μ_0, if frozen clusters are recolored independently and uniformly, then for the one-step posterior replica $\Sigma_n^{(1)}$, any algorithm τ_n, and any $\eta > 0$:

$$\lim_{n\to\infty} \mathbb{P}\left[\mathrm{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq \frac{1}{K} + \frac{K-1}{K}(\theta^{\max} + \eta)\right] = 0$$

where $\theta^{\max}$ is the asymptotic fraction of nodes in giant frozen components.

Corollary (Percolation is Necessary)

If there is no percolation ($\theta^{\max} = 0$), the maximum overlap is bounded by $1/K$ (random guess): weak recovery is impossible. Hence, ***percolation is a necessary condition for weak recovery****.*

Key Intuition

Small clusters are recolored independently, washing out their contribution. Only giant (macroscopic) components can carry global community information.

Coupling & the Sankararaman–Baccelli Bound

One Step of Classical Edge-by-Edge Swendsen–Wang

Let's run one step of standard SWENDSEN–WANG (edges) initialized at the hidden truth Σ. For any edge e, the probability that it is frozen is:

- **Case 1** $(\Sigma_i = \Sigma_j)$:

$$\mathbb{P}(\text{frozen}) = f_{\text{in}}(r_e)\left(1 - e^{-|W_e|}\right) = f_{\text{in}}(r_e)\left(1 - \frac{f_{\text{out}}(r_e)}{f_{\text{in}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

- **Case 2** $(\Sigma_i \neq \Sigma_j)$:

$$\mathbb{P}(\text{frozen}) = (1 - f_{\text{out}}(r_e))\left(1 - e^{-|W_e|}\right) = (1 - f_{\text{out}}(r_e))\left(1 - \frac{1 - f_{\text{in}}(r_e)}{1 - f_{\text{out}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

Corollary (Retrieving the S&B Bound [1])

*Under the classical edge SW dynamics, the frozen graph is **exactly** an independent bond percolation with edge probability:*

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$

Strict Improvement of the S&B bound (on the triangular grid)

Homogeneous Model ($f_{\mathrm{in}} = p$, $f_{\mathrm{out}} = 1 - p$)

On the triangular grid, we select a family of independent white triangles (like a chessboard).

- **Edge Dynamics (Standard SW)**: Frozen graph is independent edge percolation with parameter $p_{\mathrm{edge}} = 2p - 1$. Critical threshold:

$$p_c^{\mathrm{edge}} = \frac{1}{2} + \sin\left(\frac{\pi}{18}\right) \approx 0.673\ldots$$

- **Triangular Dynamics**: Frozen graph yields correlated triangle percolation. Critical threshold:

$$p_c^{\triangle} = \frac{7}{4} - \frac{\sqrt{17}}{4} \approx 0.7192\ldots$$

Corollary (Strictly Better Impossibility Bound [10])

Since $p_c^{\mathrm{edge}} < p_c^{\triangle}$, *for any* $p \in [0.673\ldots, 0.7192\ldots)$, *the standard S&B bound cannot conclude (edge dynamics percolate), whereas triangular dynamics do not, proving that weak recovery is impossible.*

Empirical Results

04
Open Questions & Conclusion

Open Questions

Limits of the One-Replica Coupling

- Our Bayesian framework combined with MCMC cluster dynamics allows us to construct **one** replica $\Sigma_n^{(1)}$.
- This only yields a bound (a **necessary condition** for weak recovery).

Towards a Necessary and Sufficient Condition

- To achieve equality (a **necessary and sufficient condition**), a framework involving **two replicas** would be required.
- How can we define and analyze such a two-replica coupling under MCMC dynamics?

Towards a Practical Community Detection Algorithm

Limits of Direct Gibbs Sampling

- Empirically, recovering communities by sampling directly from the Gibbs distribution does not perform well.
- This is likely because the **mixing time** of the Markov chain is too large.

Estimating the Correlation Matrix

- Constructing an estimator of the correlation matrix:

$$\mathbb{P}[\mathbb{I}_{\{\sigma_i=\sigma_j\}}]$$

  gives significantly better results.
- **Future work**: It would be highly relevant to theoretically study why this approach is much more robust.

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Merci.